Narendra Sir
Biology Lecturer | Founder & CEO

TOPIC 7: MUTATIONS & PEDIGREE ANALYSIS

PU-II Principles of Inheritance and Variation • Topic-Wise Worksheet

Student Name: ____ **Roll / Reg. No:** ____________ **Date / Batch:** _________

SECTION A: Multiple Choice Questions (1 Mark Each)

1. A point mutation involving a single base substitution GAG → GUG causes:

(A) Thalassemia
(B) Down's syndrome
(C) Sickle cell anaemia
(D) Turner's syndrome

2. Nitrous acid acts as a chemical mutagen by causing:

(A) Thymine dimerization
(B) Deamination of cytosine to uracil
(C) Frameshift deletions
(D) Chromosome breakage

3. In a pedigree chart, a square symbol represents a:

(A) Female
(B) Male
(C) Sex unspecified
(D) Carrier female

4. Which of the following is a classic autosomal dominant Mendelian genetic trait in humans?

(A) Phenylketonuria
(B) Myotonic dystrophy
(C) Cystic fibrosis
(D) Albinism

SECTION B: Fill in the Blanks (1 Mark Each)

5. A point mutation occurs due to alteration in a single .... pair of DNA.

6. Mating between genetically related individuals shown by double lines in pedigree is termed .... mating.

7. Physical agents that induce mutations like UV and gamma rays are called ....

8. The initial affected family member serving as the starting point for pedigree investigation is the ....

SECTION C: Very Short Answer Questions (2 Marks Each)

9. **[2M]** Differentiate between Point mutation and Frameshift mutation with examples.

10. **[2M]** Draw the standard pedigree symbols for: (a) Affected female (b) Consanguineous mating.

SECTION D: Short Answer Questions (3 Marks Each)

11. **[3M]** State three key characteristics of Autosomal Dominant pedigree inheritance.

12. **[3M]** Classify mutagens into physical, chemical, and biological categories with one specific example for each.

SECTION E: Long Answer / Essay Questions (5 Marks Each)

13. **[5M]** What is Pedigree Analysis? Describe its significance in human genetics. Draw schematic pedigrees showing inheritance of (a) Autosomal dominant trait (b) Autosomal recessive trait.

14. **[5M]** Discuss mutations in detail: (a) Gene mutations vs Chromosomal mutations (b) Molecular basis of point mutation in sickle cell anaemia (c) Major mutagenic agents.